



IPB University
— Bogor Indonesia —

13

CLIMATE
ACTION



11

SUSTAINABLE CITIES
AND COMMUNITIES



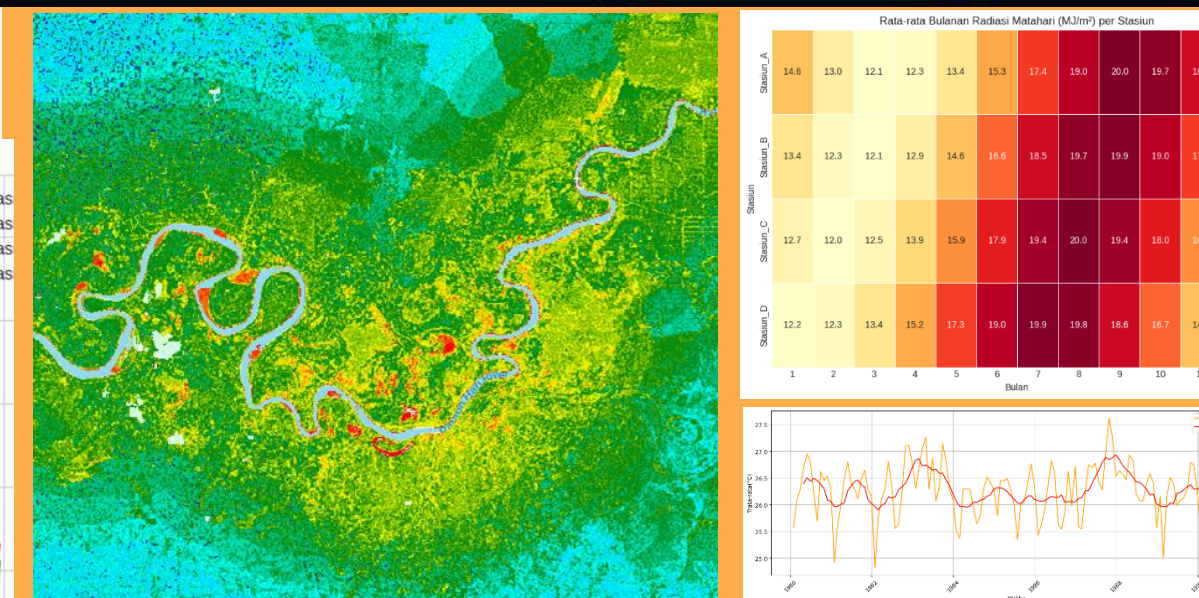
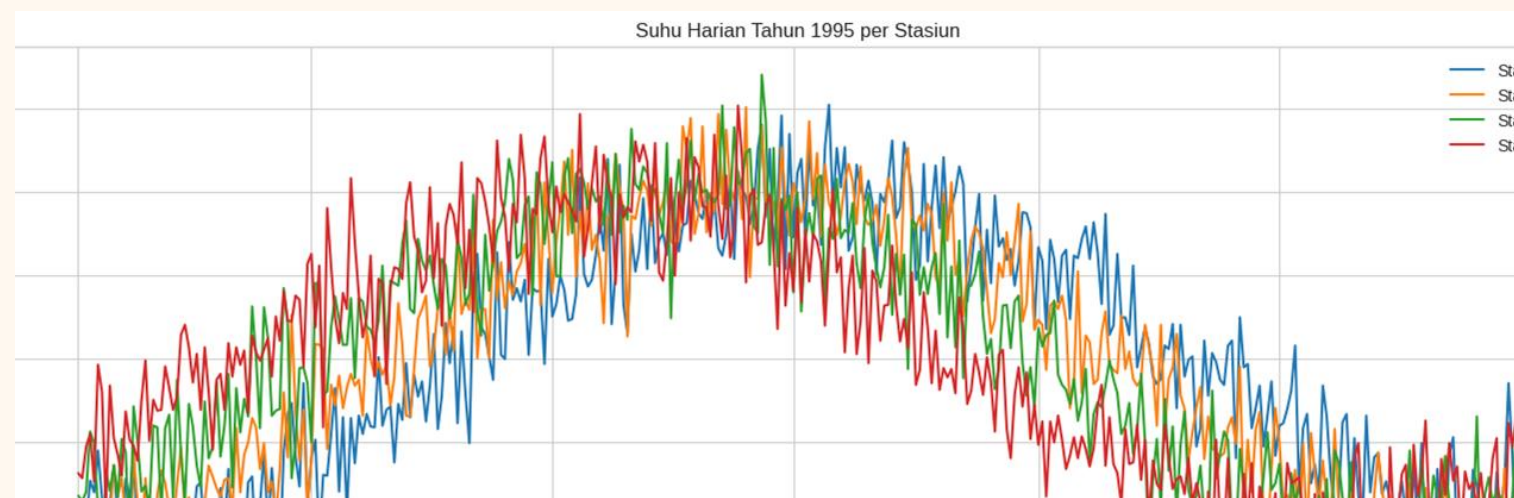
presented by
I PUTU ARIS GUNARTA

Portfolio

Social Empowerment | Climate Action

Transforming climate data into real solutions for resilient communities

[LINKEDIN.COM/IN/PUTUARIS/](https://www.linkedin.com/in/putuaris/)



Putu Aris - Climate Analyst

"Transforming climate data into real solutions for resilient communities."



As someone who once stood before a panel of global delegates and earned the **Best Position Paper on Climate Action**, I've always believed that knowledge should lead to impact.

With a background in **Applied Climatology** and a proven track record in **climate advocacy, disaster literacy training, and environmental education**, I've worked across schools, communities, and international platforms to make climate science more relatable and actionable. Through initiatives like the **Climate Vlogger Workshop**, the **Rinjani Geopark Training of Trainers**, **Internet of Things research assistant** and contributions to **zero-waste city planning**, I combine a strong scientific foundation with the **communication and leadership skills** needed to engage, inspire, and drive change.

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Related Project

AI-Based Climate Modeling

Daily Temperature Prediction Using FeedForward Neural Network

Institution: IPB University – Master's in Applied Climatology

Year: 2025

Objective: To develop a data-driven model to predict surface temperature using a FeedForward Neural Network (FNN) based on 30 years of simulated weather data from four stations (1995–2024).



Approach:

- Data preprocessing: seasonal pattern identification, normalization, and lag-feature generation
- Model building: FNN using Keras, hyperparameter tuning with Keras Tuner
- Cross-variable analysis: correlation between temperature, rainfall, humidity, and solar radiation
- Evaluation: Pearson correlation ($R = 0.97$), Taylor diagram, MAE & RMSE metrics

Key Results:

- High prediction accuracy ($R = 0.97$)
- Clear seasonal trends and climatic shifts identified through moving averages
- Demonstrated the effectiveness of ANN in modeling nonlinear climate systems
- Model supports decision-making in agriculture, disaster preparedness, and local adaptation

Tools & Skills:

Python, Keras, NumPy, Pandas, Matplotlib, Climate Data Analytics, Neural Network Tuning, Time Series Analysis

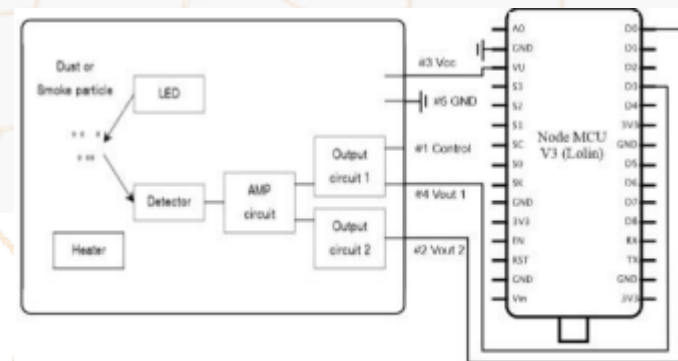
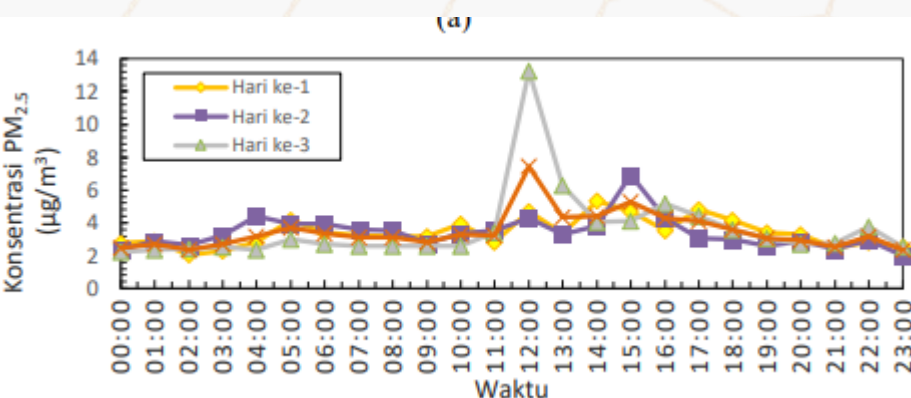
Related Project

Prototype of Low Cost Portable PM2.5 Dust Monitoring Tool with Data Logging and Web Visualization Sensor Using ESP32

Role: Research Assistant—Climatology Lab, IPB

Year: 2018-2019

Designed and developed an integrated IoT system to monitor PM2.5 dust levels using the **DSM501** sensor and **ESP32**. The system includes **local data storage via microSD** and a **local web interface (HTML/PHP)** that enables users to **sort, filter, download, and visualize the data** directly.



System Architecture:

Sensor: DSM501 PM2.5 Dust Sensor

Microcontroller: ESP32

Data Storage: microSD Module + Localhost SQL Database

Interface: HTML/PHP-based UI for data interaction

Features:

- Real-time PM2.5 monitoring
- Web dashboard with sorting, download, and visualization options
- Local SQL data storage for offline access and analysis

Skills Gained:

IoT prototyping, embedded programming (Arduino IDE), database design (SQL), local server deployment (XAMPP/LAMP), HTML/CSS/JS for dashboard UI, serial data parsing, and environmental sensor integration.

Potential Applications:

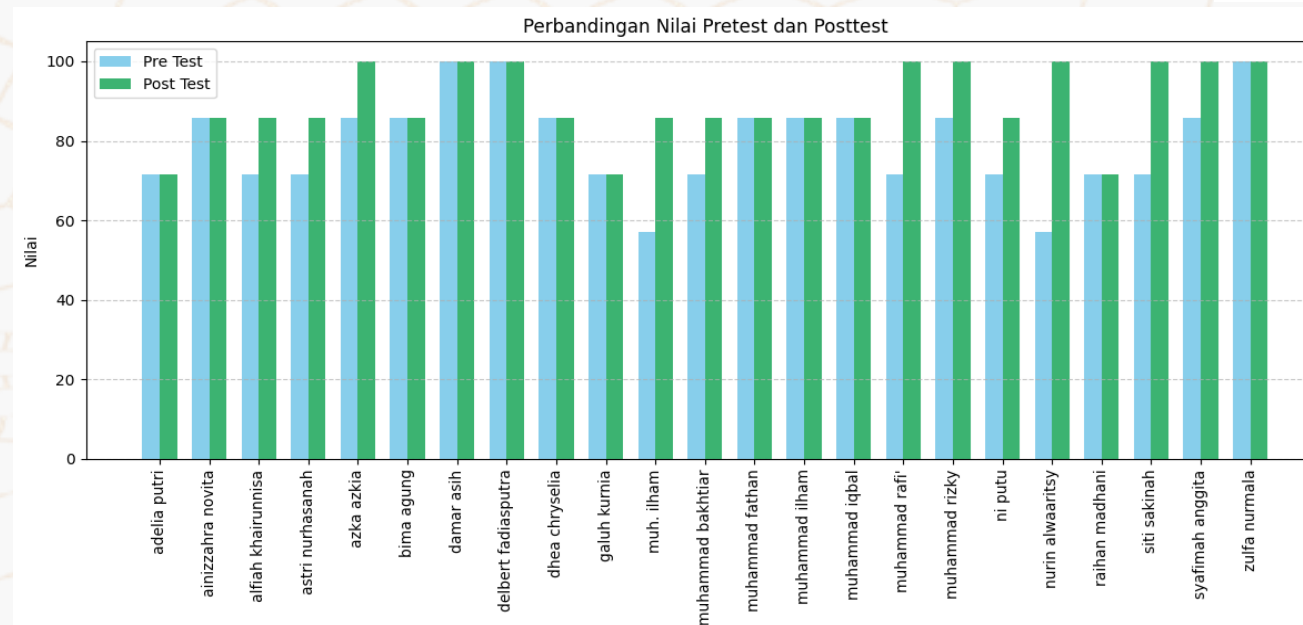
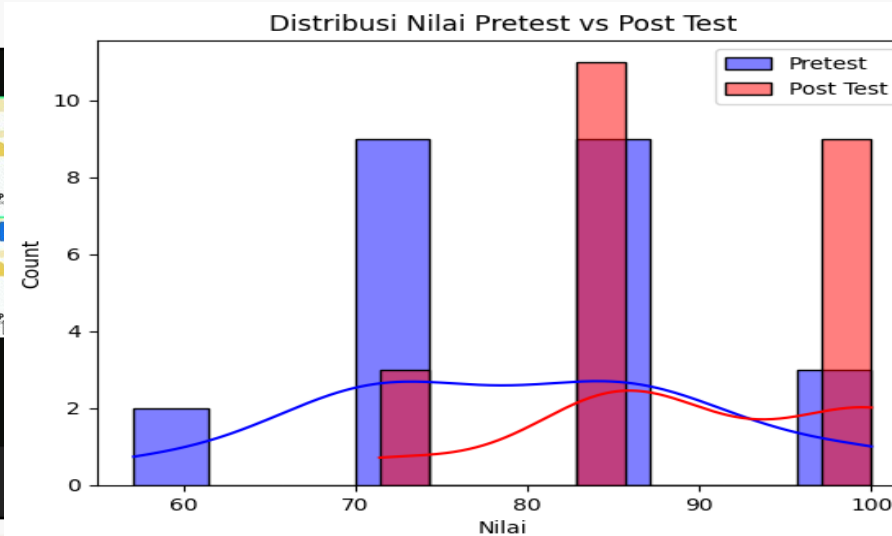
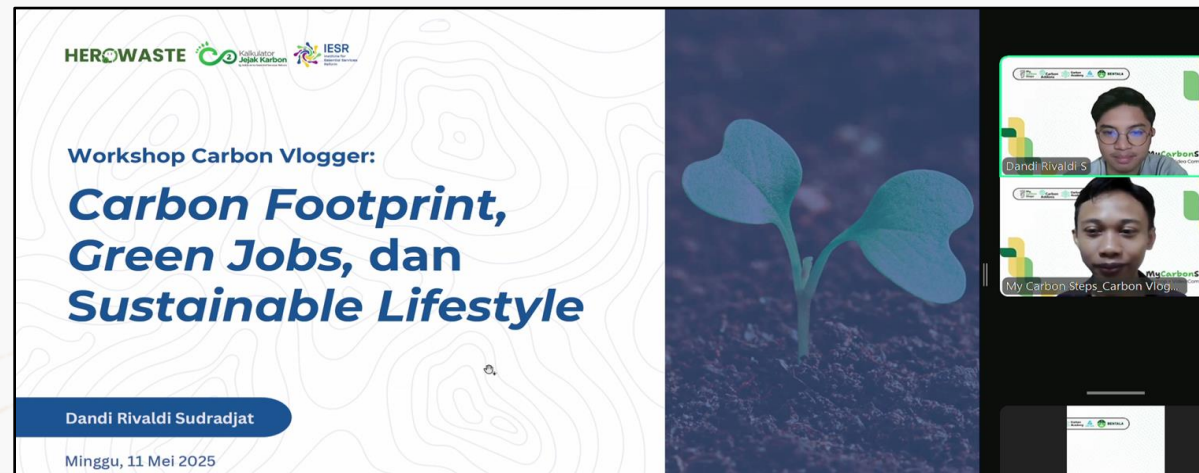
This system can be deployed for **community-based air quality monitoring**, urban pollution mapping, or citizen science platforms, especially in under-monitored or high-risk environments.

Related Project

Social Project: Carbon Vlogger Workshop & Challenge

Role: Project Leader(Carbon Academy Vol. 2)

Year: 2025 | **Scale:** National (Virtual)



Key Achievements:

- Secured 2.5 million IDR sponsor funding
- 35 participants from multiple provinces across Indonesia
- Notable improvement in post-test results (knowledge gain evaluation)
- 13 environmental vlogs successfully produced and published promoted on Instagram.
- 77% of participants completed both pre- and post-tests, showing significant growth in climate awareness

My Contributions:

- Designed training flow and educational content
- Developed evaluation methods (pre- and post-tests including qualitative assessment)
- Created a social media dissemination strategy
- Integrated “green jobs” and “climate storytelling” themes in content
- Submitted and accepted on International Conference WRI 2025

[Instagram.com/mycarbonsteps](https://www.instagram.com/mycarbonsteps)

Empowering Youth Climate Action through Digital Engagement

Integrating Climate Resilience into Community-Based Learning and Local Development Planning

**I Putu Arisgunarta^{1.3},
Arrozaq Ave^{1.2}, Nasya Augusta^{1.4}, Qory Salsabilla^{1.5},
Ni Putu Ayu Eka Sundari^{1.6}**

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³Department of Applied Climatology, Faculty of Mathematics and Natural Sciences, Graduate School, IPB University,
Bogor, Indonesia

⁴Department of Digital Business, Faculty of Economics and Business Education, Universitas Pendidikan Indonesia,
Bandung, Indonesia

⁵Department of Sociology, Faculty of Social Sciences and Law, Universitas Negeri Jakarta, Jakarta, Indonesia

⁶World Resource Institute (WRI), Jakarta, Indonesia

Latar Belakang

When Might Temperature Thresholds Be Reached?



Notes: *SSP1-1.9 **SSP5-8.5

Source: IPCC,
21.08.05

WORLD RESOURCES INSTITUTE



<https://www.shutterstock.com/image-vector/set-young-characters-phones-teens-260nw-1960376221.jpg>
https://pngtree.com/freepng/youth-marching-for-climate-action_22303969.html

- **Perubahan iklim:** tantangan global dengan dampak nyata di berbagai sektor
- **Tingginya emisi karbon:** akibat aktivitas manusia, butuh mitigasi dan adaptasi

Anak muda → Tingkat konektivitas digital tinggi (BPS, 2023)

- 94% menggunakan internet
- 96% memiliki smartphone

49,7% remaja Indonesia memiliki literasi iklim yang rendah (Martha et al., 2025)

Kesenjangan antara kesadaran dan aksi (Moser, 2016; Chen et al., 2025)

Tujuan

1. Merancang kegiatan interaktif yang menantang generasi muda untuk memahami pentingnya jejak karbon
2. Mengenalkan konsep jejak karbon dalam aktivitas sehari-hari untuk meningkatkan kesadaran lingkungan

Kebermanfaatan Proyek

Meningkatkan kesadaran dan aksi pemuda terhadap jejak karbon serta isu lingkungan.

Mengembangkan keterampilan digital dan komunikasi lewat pembuatan vlog.



Memanfaatkan platform interaksi maya (sosial media) generasi muda untuk menyebarkan kesadaran lingkungan ke jaringan sosial mereka.

Membuka wawasan generasi muda tentang peluang karir green jobs.

Methodology

Our Approach

-  **1** AIDA Model
Digital engagement (Attention, Interest, Desire, Action)
-  **2** Participants
30 youth (age 17-25) from 10 provinces

Data Analysis Method



Teori AIDA sebagai Katalis

Pendekatan proyek ini sejalan dengan teori AIDA (Attention, Interest, Desire, Action):

1. Attention (Perhatian)

- *Carbon vlogger* menarik perhatian anak muda melalui media digital yang mereka gunakan sehari-hari.

2. Interest (Minat)

- Dengan menyajikan konten menarik serta program workshop & challenge, tentang perubahan iklim dan jejak karbon, program ini menumbuhkan rasa ingin tahu mereka.

3. Desire (Keinginan)

- Pemahami bahwa mereka dapat berkontribusi dalam mengurangi emisi karbon dan bahkan menjadikannya peluang karir di bidang *green job* meningkatkan minat anak muda untuk mengambil aksi nyata. Sistem rewards juga menjadi katalis untuk meningkatkan minat anak muda

4. Action (Tindakan)

- Peserta didorong untuk membuat vlog edukatif, menyebarkan informasi, serta menginspirasi orang lain untuk mengambil tindakan nyata dalam mitigasi perubahan iklim.

The Innovation: Carbon Vlogger Challenge 2025

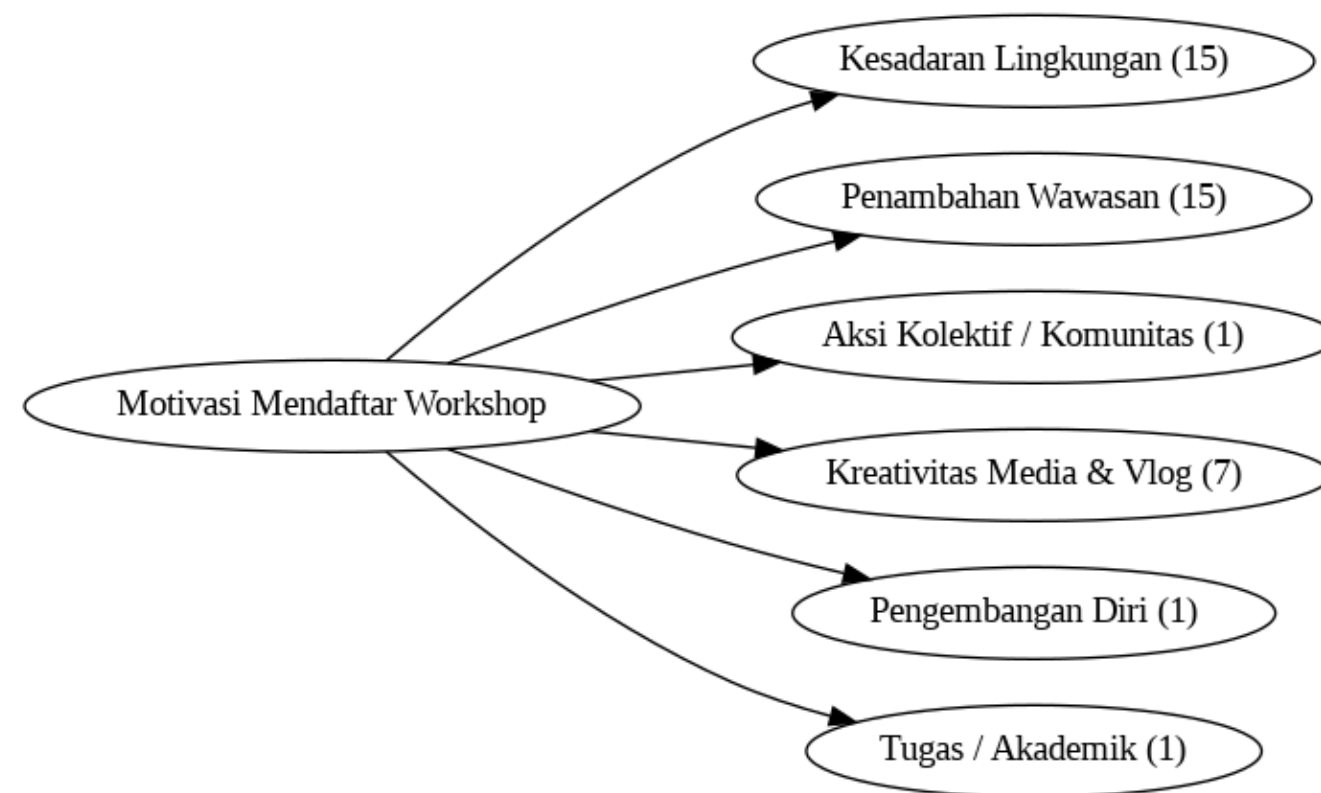


**Hybrid model:
Climate education workshop + digital storytelling**

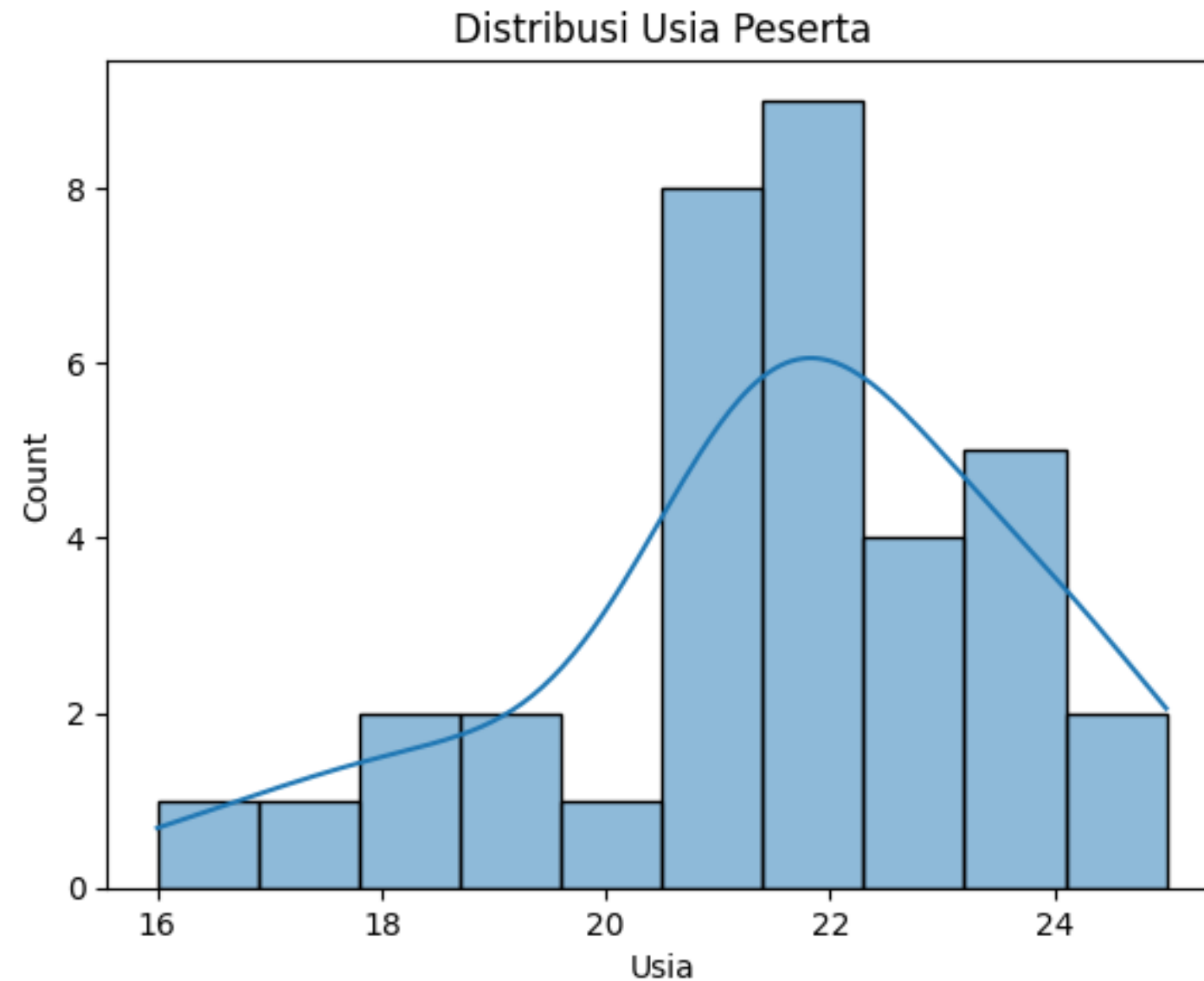
- Online workshop on carbon awareness & video creation
- Youth-led vlog production challenge
- Peer-to-peer knowledge sharing at scale
- Low-cost, replicable model for local governments

Bottom-up approach using creative communication as both educational tool and catalyst for change

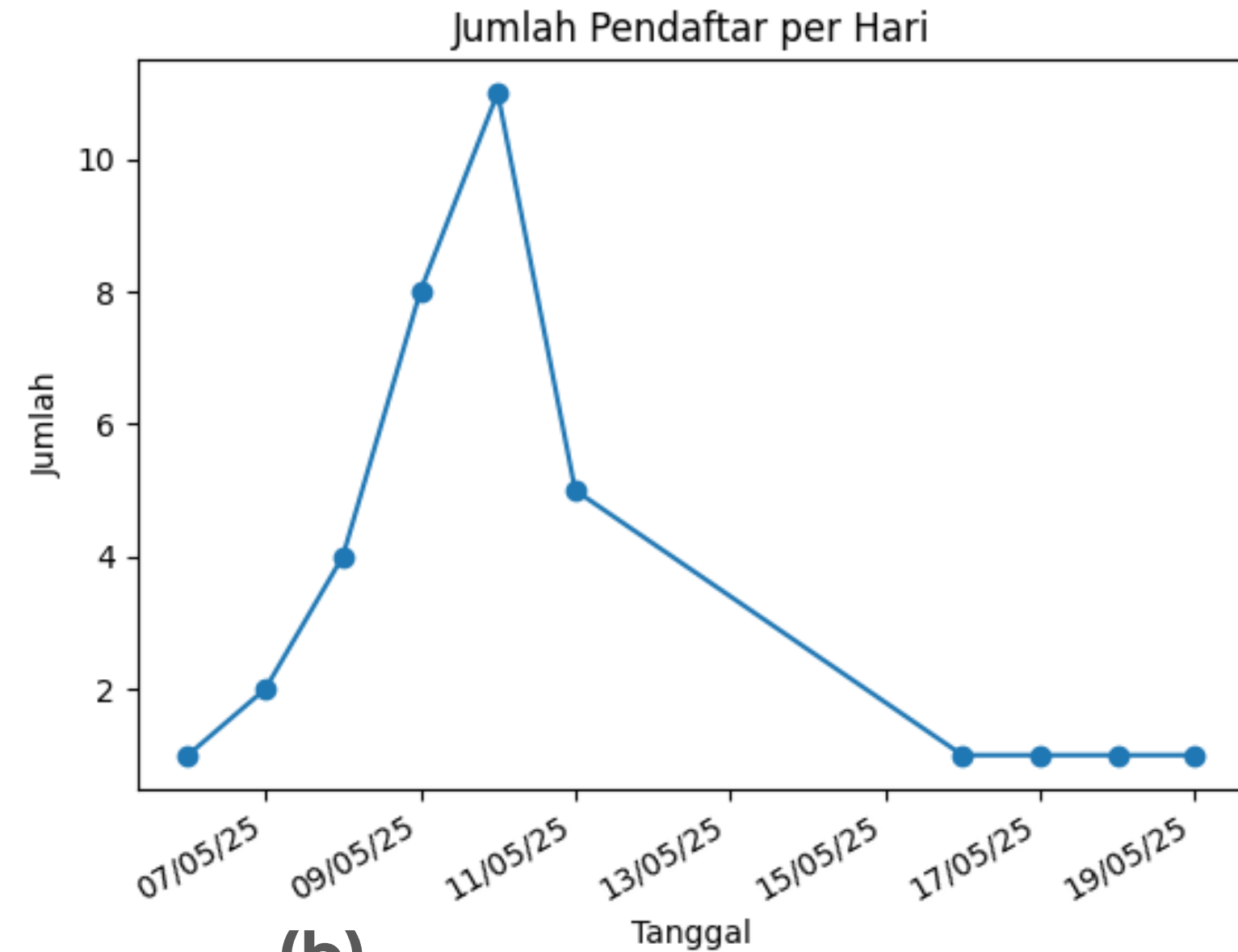
Apa motivasi peserta?



Realisasi Proyek



(a)



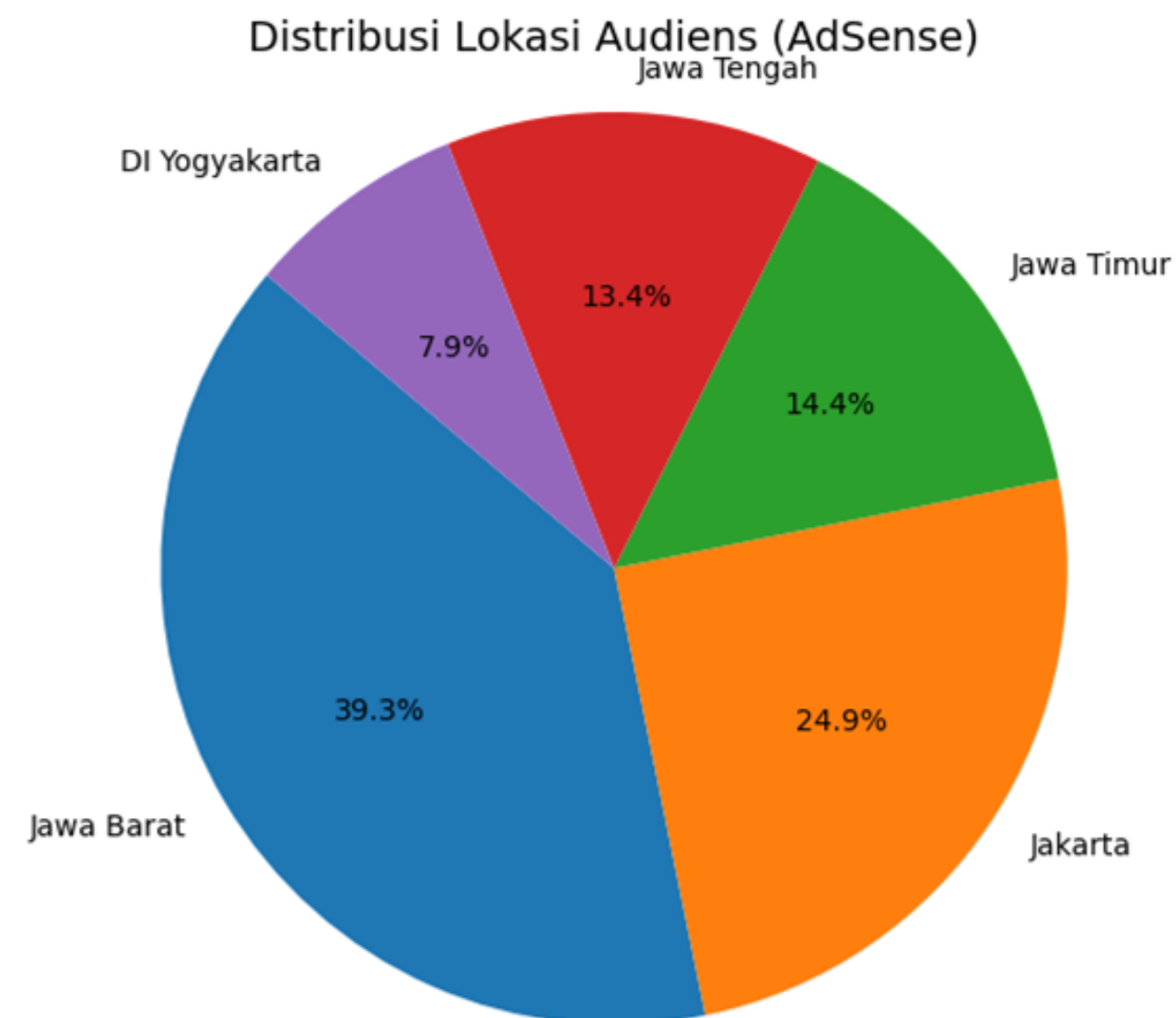
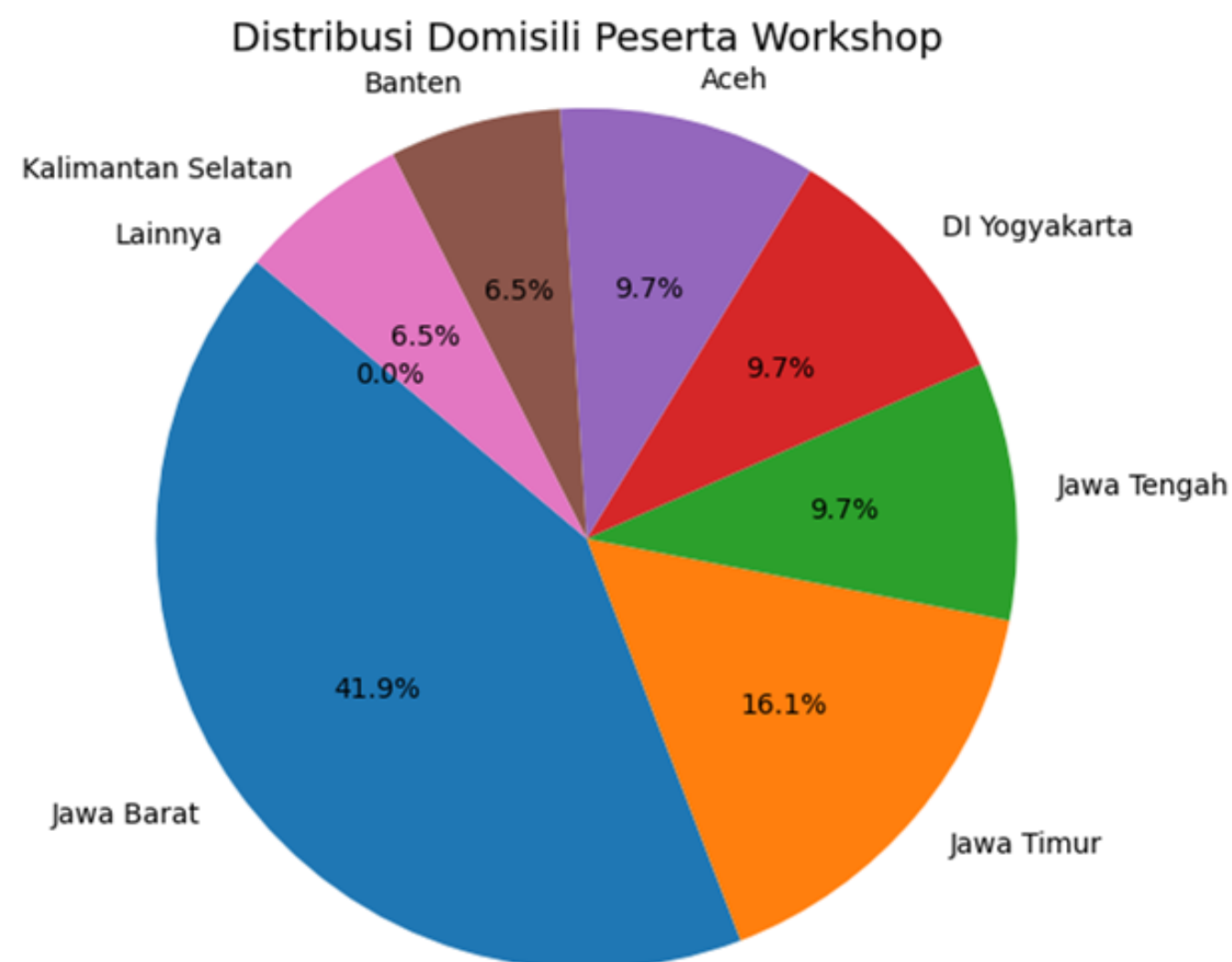
(b)

- (a) Rata-rata peserta berusia 20 tahunan,
(b) Terjadi peningkatan pendaftar saat menggunakan adsense di tanggal 9-11

Pendaftar: **65.7% Wanita**

Realisasi Proyek

Distribusi peserta mirip dengan lokasi audiense instagram



Instagram merupakan platform yang sangat efektif untuk menjangkau generasi muda

Outcome: Knowledge Increase

Ceiling Effect on Pre-Post Test Assessment

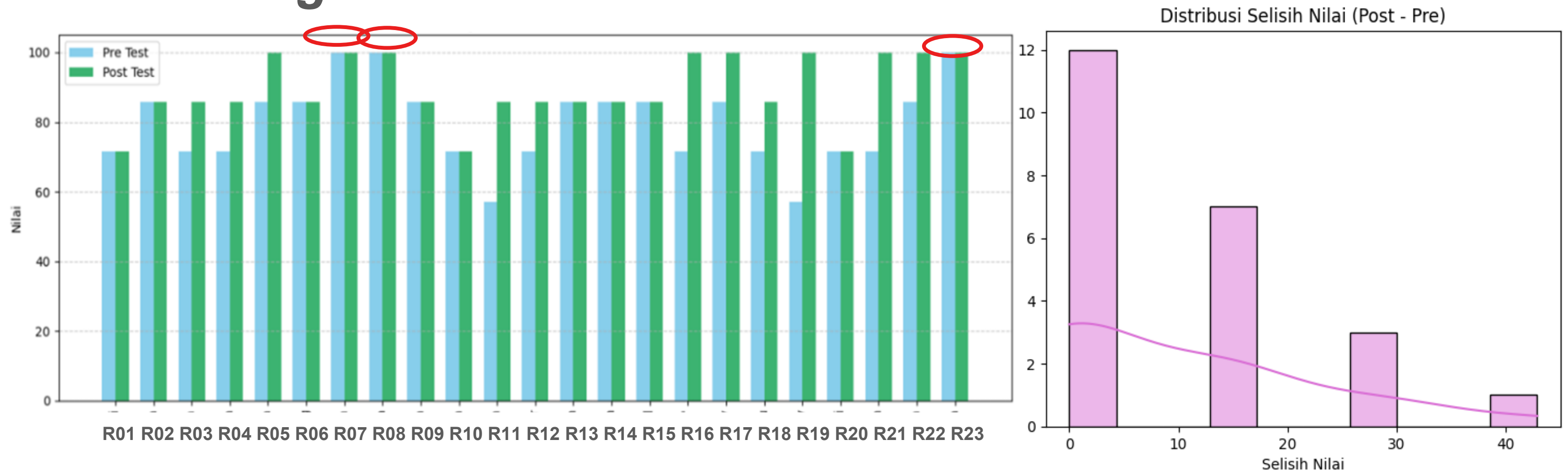


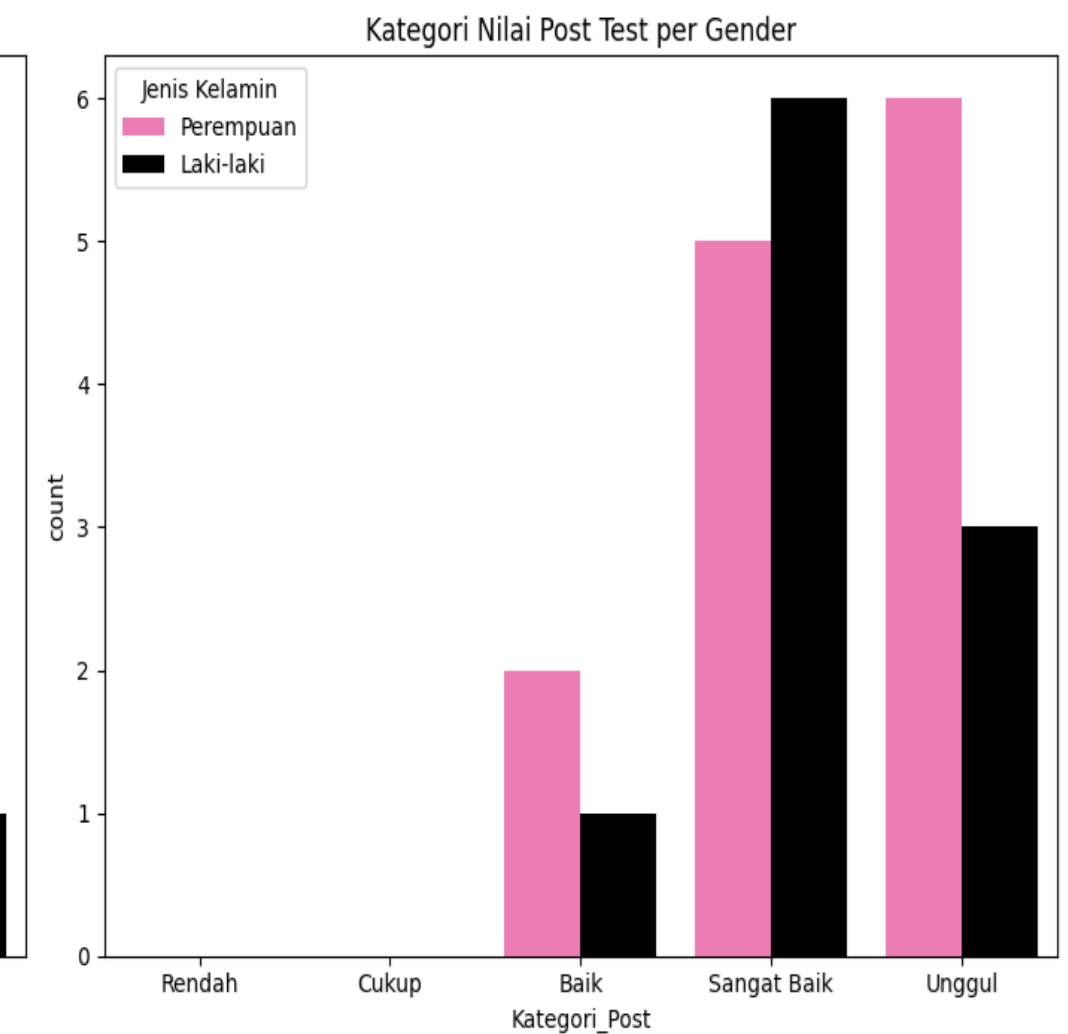
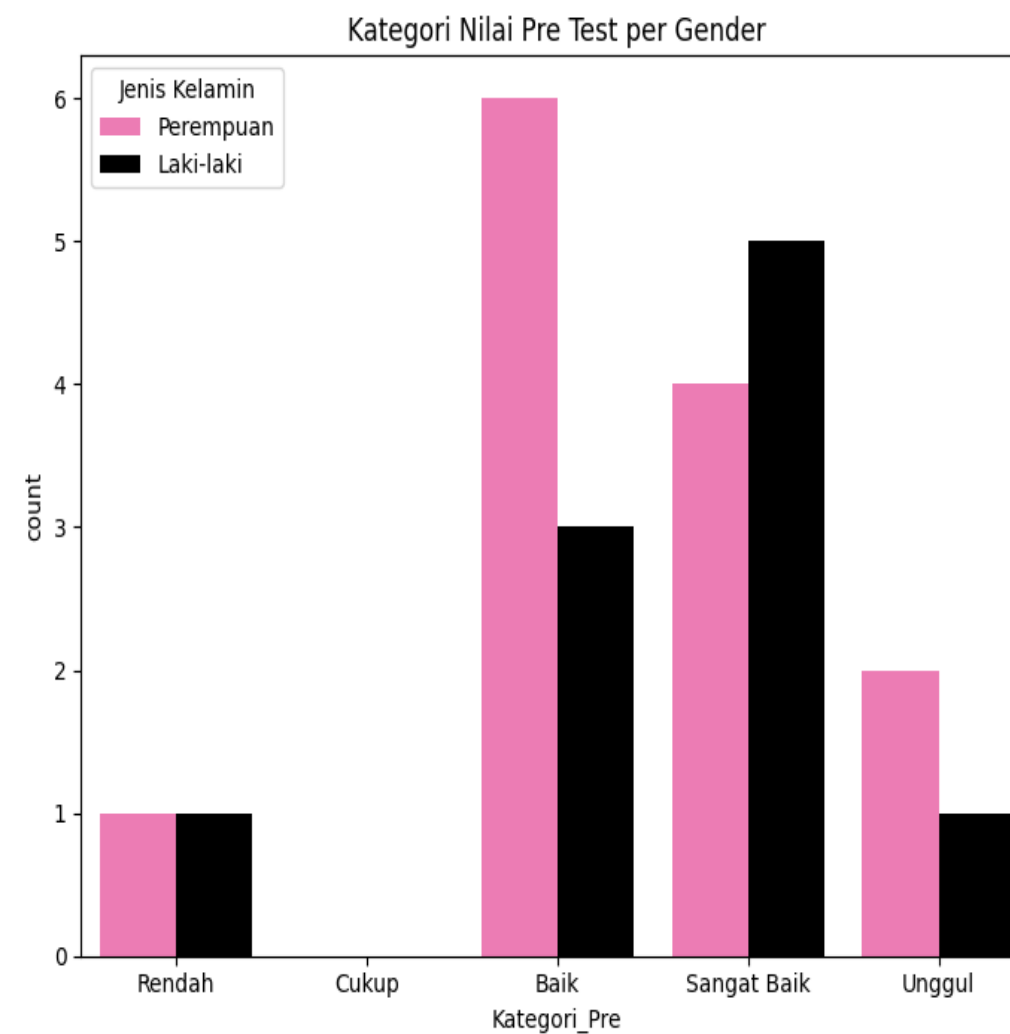
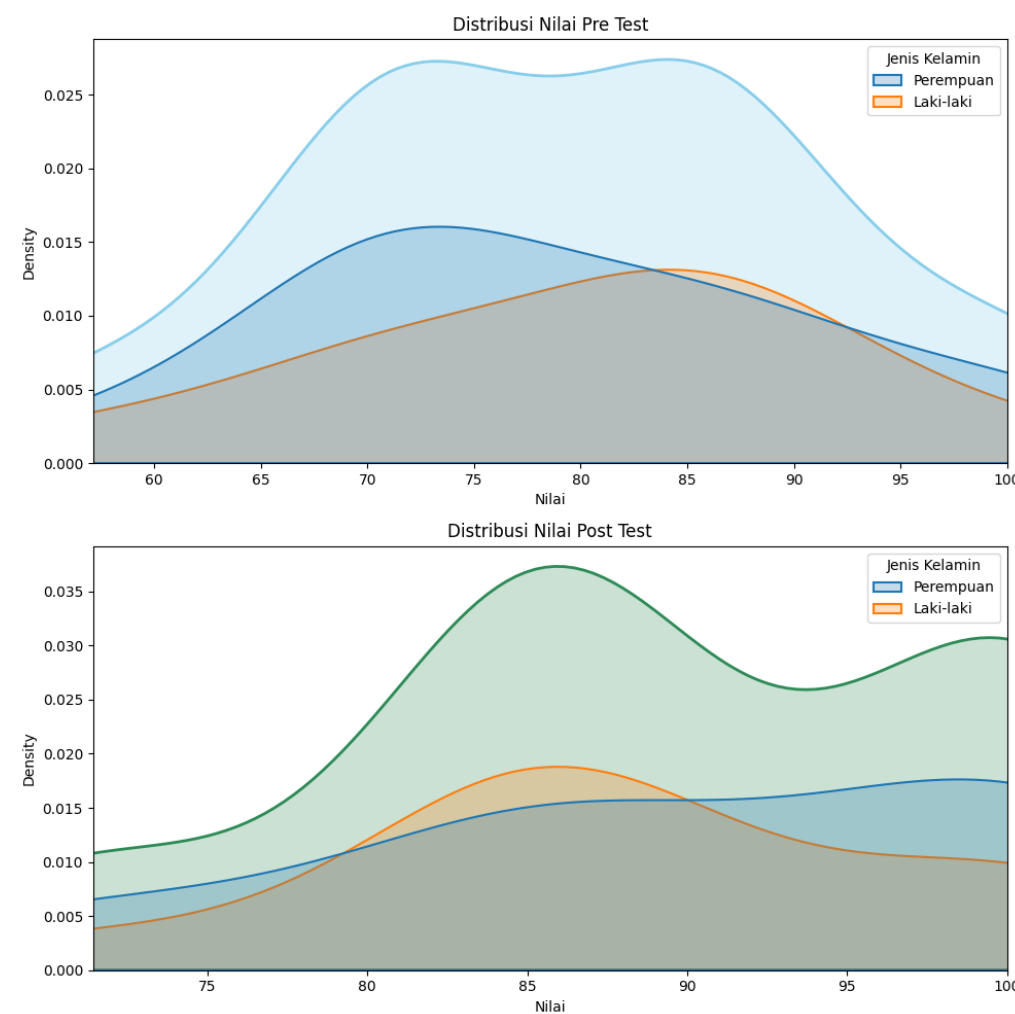
Figure 1. Pre-Test and Post-Test Results (left: blue color indicate pre test, green indicate post test) and Score Distribution (right: purple graph)

Terjadi peningkatan skor pre-test dan post test peserta (**52,17% tetap, 47,83%**

Ceiling Effect

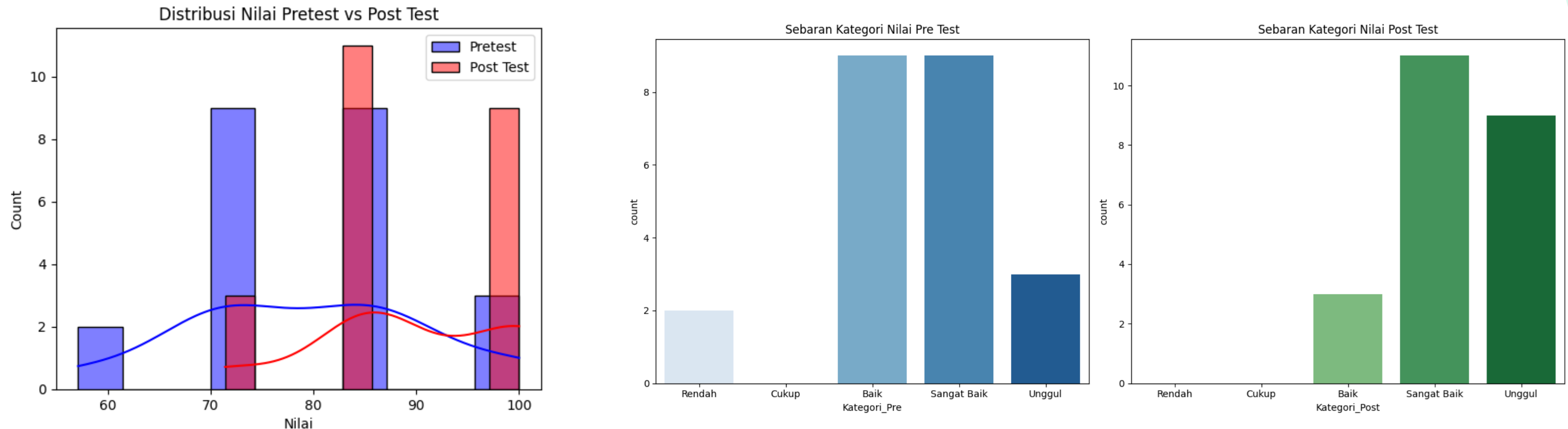
Beberapa peserta memperoleh nilai sangat tinggi pada pre-test dan post-test, sehingga peningkatan skor terlihat kecil meskipun pemahaman mereka sebenarnya sudah kuat sejak awal.

Outcome: Knowledge Increase



Data menunjukkan peserta **wanita memiliki** rata-rata **skor yang lebih tinggi** dibandingkan peserta laki-laki

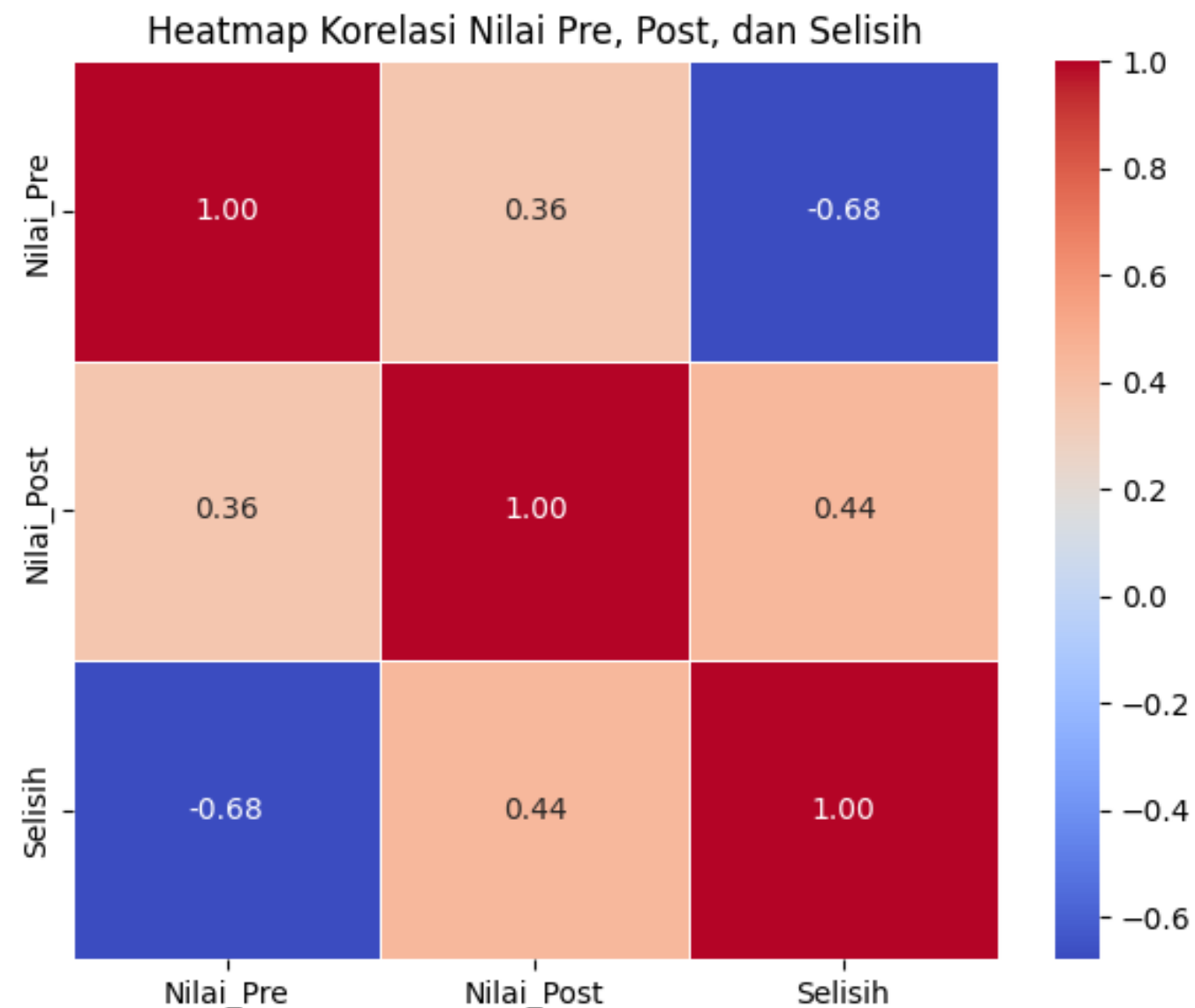
Outcome: Knowledge Increase



Pergeseran nilai peserta dari kategori **rendah-unggul**, dengan median di baik-sangat baik, menjadi **baik-unggul**, dengan median di sangat baik-unggul.

(kategori rendah < 60, baik 60-70, sangat baik 80-90, unggul > 90)

Outcome: Knowledge Increase



Heatmap dapat menjadi indikasi bahwa **intervensi lebih berdampak pada peserta dengan kemampuan awal yang lebih rendah ($R=-0.68$).**

Outcome

Knowledge Increase

14.7%

Average Knowledge Increase
in carbon footprint
understanding

77%

Workshop Participation
(Pre-test & Post Test)

0.38

N-Gain Score
Medium effectiveness
(Hake 1998; Colleta &
Steinert 2020)

13 Vlogs

Educational Vlogs
youth-created videos
(**56.7%** from total participants)

Outcome

Digital Outreach Impact

20.000+

Total Vlog Views



30.000+ Impressions

133 Profile Visits

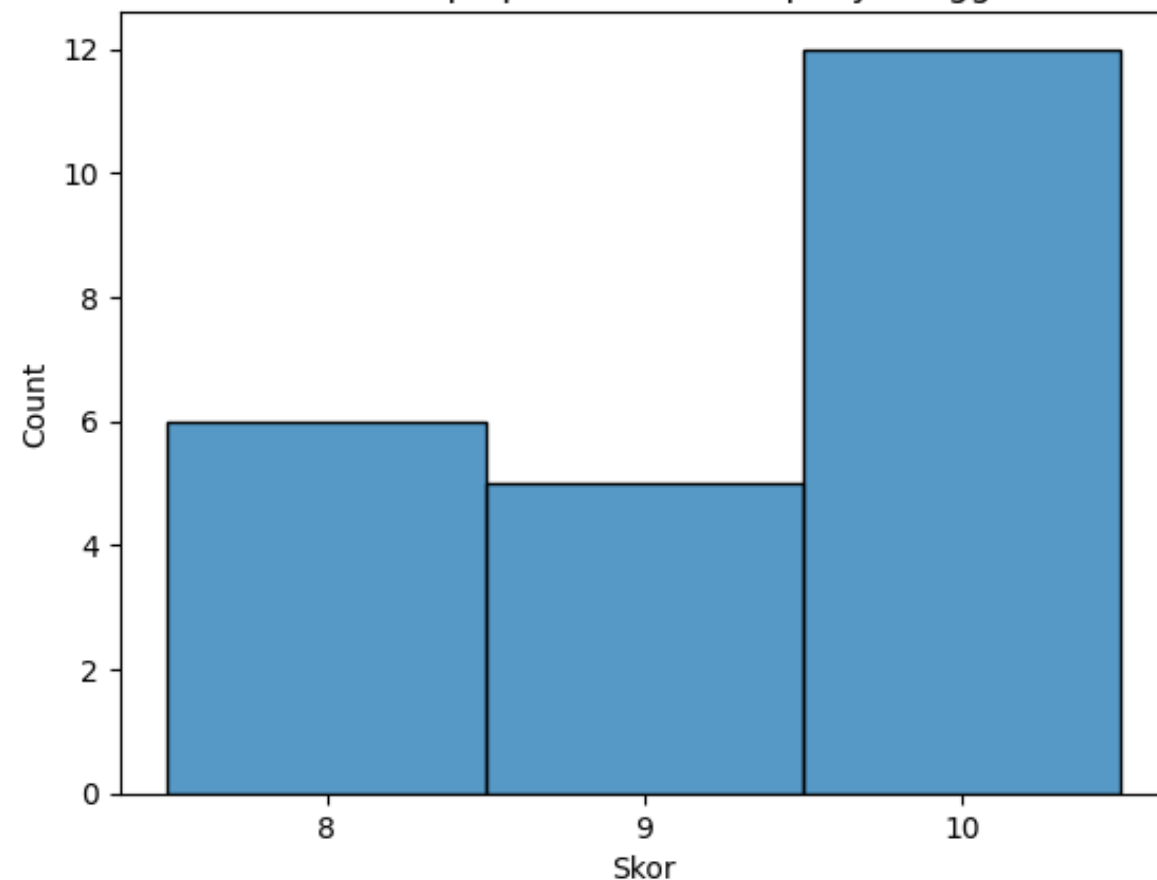
25 New Followers

IDR 50.000 Total Ad Spend

IDR 1,400,000 Rewards

Apa kata peserta?

Distribusi Penilaian: Dari 1-10, berapa penilaian untuk penyelenggaraan secara keseluruhan?



Pengurangan emisi karbon dan jejak karbon sebagai aksi nyata gaya hidup sehat yang berkelanjutan, serta peluang berkarir di *Green Jobs*

Kesimpulan

“Anak muda sebagai pilar literasi iklim digital murah, inklusif, dan scalable”

Key Findings:

- Keterlibatan digital anak muda (RQ1): terbukti meningkatkan literasi jejak karbon melalui pembelajaran interaktif, workshop, dan *carbon vlogger challenge*.
- Pendekatan aksi iklim berbiaya rendah (RQ2): menyediakan jalur yang mudah direplikasi bagi komunitas dan pemerintah daerah untuk mengintegrasikan komunikasi adaptasi iklim ke dalam

“Digital engagement is not just about technology—it's about empowering voices”

Conclusion

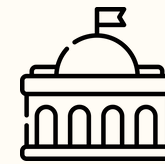
Recommendation

for academia:

1. Mainstream youth-centered digital climate learning models
2. Strengthen climate literacy initiatives

for private sector

1. Adopt and include youth as a pillar of low-cost digital climate awareness model CSR or ESG
2. Expand opportunities for green jobs in communication roles



for government

1. Align RAN-API with replicable low-cost digital engagement model
2. Build climate resilience through both infrastructure AND social innovation



Thank You

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#CarbonVlogger #ClimateResilience
#YouthAction



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